

2 Bouncing back

The sound waves reflect, or bounce, off the ceiling and walls of the cave, creating echoes.

3 Bouncing around

In a cave, there are lots of hard surfaces, all of which reflect your shout. Your shout might even bounce more than once before returning.

Decibel levels

Loudness is measured in decibels (dB). Sounds get louder as you get closer, so to compare loudnesses, you have to take distance into account. Here are some examples:

30 dB

Rustling leaves



50 dB

Quiet music



60 dB

Speaking



80 dB

Heavy traffic



115 dB

Baby crying



125 dB

Pneumatic drill



140 dB

Jet engine



4 Getting weaker

The sound waves start strong but get weaker as they spread out. By the time they reach your ears, they have become much weaker, making the echo sound much quieter.

Echoes don't take long to come back. Sound travels **1,100 ft (340 m) every second!**